

OPTIMIZATION OF CORE GROOVE GEOMETRY FOR THE MANUFACTURE AND OPERATION OF COMPOSITE SANDWICH STRUCTURES IN WIND TURBINE BLADES

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Abstract

The design of modern wind turbine blades has scaled up significantly in the past decades. Thus, the need to optimize design and manufacturing practices has become more important for manufacturers to maintain the quality of production and profitability in the competitive modern market. Specifically, the shallow grooves that are cut into sandwich core materials to aid Vacuum-Assisted Resin Infusion (VARI) are a particular feature with an opportunity for optimization in terms of fluid flow performance and resin mass savings. This work focuses on the effect of vacuum infusion channel geometry on fluid flow transport (quantified by an equivalent permeability), and analytically demonstrates how the optimization of aspect ratio for a rectangular cross-section can be beneficial for fluid flow. Channel flow experiments were also used to validate some of the trends observed from parametric analysis, while variations of the optimized channel designs were evaluated and compared in terms of total resin uptake.

1. Introduction

As modern wind turbine blades continue to grow in scale, there is also a significant need to reduce weight and improve manufacturing reliability for greater efficiency in operation and production. Large regions of the blade structure consist of composite sandwich panels, with a lightweight core material surrounded by non-crimp glass fiber fabric skins, manufactured by large-scale Vacuum Assisted Resin Infusion (VARI). While the main purpose of lightweight core materials is to increase the bending stiffness of the sandwich panel, many special grooves are also machined into the core materials to facilitate improved forming or infusion behavior.

The design of such features is still reliant on the empirical experience of blade manufacturers. Much of the existing composite manufacturing literature related to liquid composite molding in the presence of open channels has focused on the development of numerical models for the simulation of the infusion process. Bickerton et al. [1] discussed the effect of square channel size on flow behavior, while the motivation behind particular channel dimensions was not discussed. The simultaneous application of the core layer with square channels and a distribution medium was studied by Zhao et al. [2] in terms of filling time reduction. However, the selection of groove dimensions and spacing was not discussed. The same is true for other studies [3, 4, 5], where Control Volume Finite Element Method (CVFEM) models have been used for the prediction of flow patterns and filling times during the infusion of structures containing core layers with channels. Those models were validated with arbitrarily selected grooves, without justification of their dimensions for a given lay-up. Hence, research into the optimization of

channel dimensions and spacing has seen little work, despite the potential benefits of increased manufacturing speed, improved reliability, reductions in resin uptake, or a reduction of overall blade mass.

Ultimately, in this work, an analytical investigation of channels in sandwich core material is undertaken, focusing on the influence of their geometry on both fluid transport potential and resin uptake, with the support of channel flow experiments.

2. Analytical investigation

2.1. Equivalent permeability of channels

Considering the effect of channel geometry on flow behavior, the fluid flow velocity inside the rectangular channel can be described simply with the Hagen–Poiseuille equation [6]. Hence, the volumetric flow rate and an equivalent permeability for a rectangular cross-section, with height – H and width – W , can be determined for different boundary conditions. In the case of a core material channel on top of a permeable fabric layer, the treatment of the interface between the channel and the fabric is often a significant consideration. When all four channel walls are treated as solid and impermeable, non-slip boundaries apply (the fluid velocity at the walls is zero), and an equivalent permeability can be determined according to Eq. 1 [7]. Conversely, if the fabric is considered as a porous medium, the non-slip condition no longer holds, and the slip velocity at the boundary interface is influenced by the volumetric flow rate within the porous fabric [8]. Subsequently, the calculation of equivalent permeability becomes far more complex, as seen in Eq. 2, requiring dimensionless slip parameter α that depends on the permeable fabric structure and its permeability, K [3, 7].

$$K_{eq} = \frac{H^2}{12} \left[1 - \frac{192H}{\pi^5 W} \sum_{n=1,3,5,\dots}^{\infty} \frac{\tanh\left(\frac{i\pi W}{2H}\right)}{n^5} \right] \quad (1)$$

$$K_{eq} = \frac{1}{HW} \sum_{n=1,3,5,\dots}^{\infty} \frac{8}{n\pi\beta_n^3} \left[W - \frac{\sinh(\beta_n W)}{\beta_n} - \frac{1 - K\beta_n^2 - \frac{\sqrt{K}}{\alpha}\beta_n \sinh(\beta_n W) - \cosh(\beta_n W)}{\sinh(\beta_n W) + \frac{\sqrt{K}}{\alpha}\beta_n \cosh(\beta_n W)} \frac{\cosh(\beta_n W) - 1}{\beta_n} \right] \quad (2)$$

Where $\beta_n = \frac{n\pi}{H}$.

The dimensionless slip parameter depends on the structure of the porous medium and can be obtained by fitting an experimental flow rate distribution through the channel with experimental data [7]. Previous work has implemented $\alpha = 0.4$ for a unidirectional glass fabric with $K = 1.15 \times 10^{-9} \text{ m}^2$ and fibre volume fraction, V_f , of 19% [7], while another study mentioned that α varies between 0.8 and 1.2 for a range of permeability values and fibre volume fractions that are more representative of structural composites applications [9]. However, all these papers acknowledge that at very low flow velocities (due to low fabric permeability), the effect of the permeable interface between the core and fabric may be neglected, and the duct flow assumption of Eq. 1 is reasonable. Based on these findings from other studies, the difference between the models in Eq. 1 and Eq. 2 has been plotted against the range of $0 \leq \alpha \leq 1$ for the fabric properties used in this study (Figure 1). Here the difference between the two models is less than 2%, which aligns closely with existing literature and validates the utilization of Eq. 1, assuming duct flow with non-slip boundaries, for the calculation of the equivalent permeability of rectangular channels.

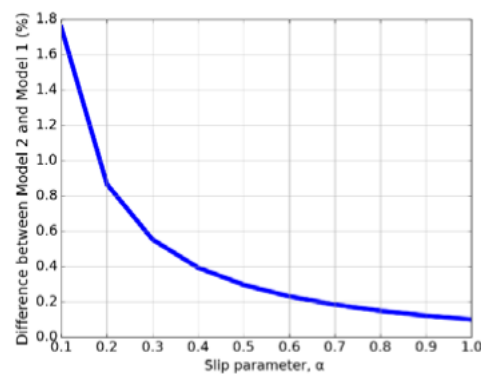


Figure 1. Difference between permeability models with and without slip velocity boundary conditions as a function of the α parameter for a fabric with $K = 1.25 \times 10^{-10} \text{ m}^2$ and $V_f = 50.6\%$.

2.2. Parametric study of channel geometry

In the context of VARI manufacturing, the selection of channel shape and dimension needs to be considered. For that reason, a parametric study of several geometries has been performed in terms of their equivalent permeability and cross-sectional area. A number of equivalent permeability solutions can be derived, from known volumetric flow rate equations [6], for simple shapes and compared relative to cross-sectional area. Figure 2 depicts the equivalent permeability of a range of channel shapes: circle, square, and rectangular cross-sections with aspect ratios (AR) 2, 3, and 10. The circular cross-section has the most effective equivalent permeability due to a uniform distribution of flow velocity and pressure gradient. Circular channels can be manufactured through the thickness of core material but cannot be employed to support in-plane flow behavior due to the lack of a potential flat interface with the adjacent fabric. Instead, rectangular channels are commonly used in industrial applications for this purpose.

Square channels have slightly lower performance in terms of equivalent permeability compared to circular channels. The equivalent permeability decreases for rectangular shapes with the increase of AR for a given cross-sectional area. The equivalent permeability value for any rectangular cross-section with a fixed area will always lie between two bounds: the upper bound is that of the square cross-section ($AR = 1$), with lower bounds for the cases where AR tends to infinity or zero. For example, in the case of a 10 mm^2 cross-sectional area, a square channel ($AR = 1$) has an equivalent permeability 4.5 times higher than that of a high-aspect ratio ($AR = 10$) rectangular channel.

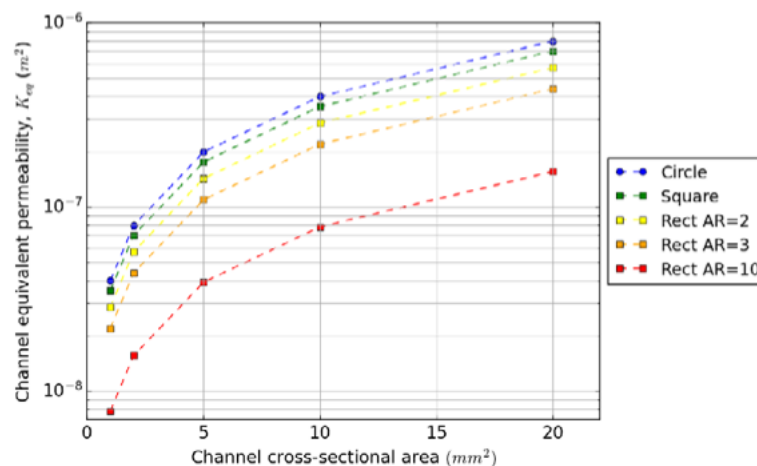


Figure 2. Equivalent permeability of channels with different geometries and cross-sectional areas.

3. Channel flow experiments

3.1. Materials and methods

In order to validate the predicted influence of cross-section geometry on the fluid transport performance of different channels, a set of representative flow experiments were performed. Four layers of 250 mm × 900 mm biaxial glass fabric, for a total thickness of 2.2 mm, were sandwiched between a glass mold and a polymethyl methacrylate (PMMA) plate with a single specific channel geometry machined into it. The 250 mm × 900 mm × 14.7 mm transparent acrylic plate was chosen instead of the actual core material in order to film the fluid flow behavior through the 900 mm long channel. A 200 mm peel-ply breaking zone was used to slow the flow progress, once the fluid reached the end of the channel. This whole configuration was placed under a vacuum bag, with the flow being driven by an inlet fluid reservoir sitting in ambient conditions and an outlet vacuum pump set to 7500 Pa. An 85% glycerol-water solution, representative of an infusible resin, was used as the test fluid, with a viscosity of 85 cP at 22 °C.

Three different channel designs were considered in this study:

- Case A: 1.5 mm × 3 mm, representative of a typical design used in the wind industry that was treated as the “reference”.
- Case B: 1 mm × 3 mm, demonstrating the effect of a reduction in channel width from the reference.
- Case C: 1.5 mm × 6 mm, demonstrating the effect of an increase in channel depth from the reference.

Flow experiments for each channel design were repeated three times. The experimental setup is presented in Figure 3.

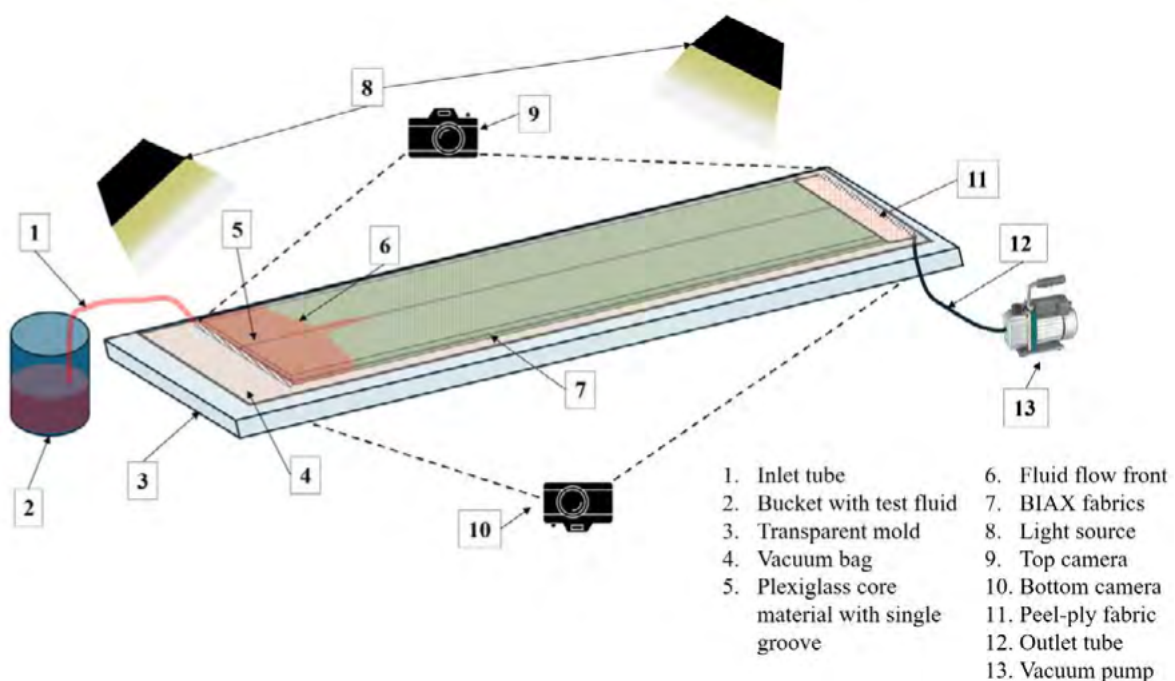


Figure 3. Setup for the channel flow experiments.

3.2. Flow front tracking

Videos from the flow experiments were processed with a Python image-processing script based on the OpenCV library. The flow front position was detected based on the difference between two frames. The obtained flow front was visualized in the recorded video (Figure 4), and stored as a set of data points, each with corresponding coordinates and timestamp.

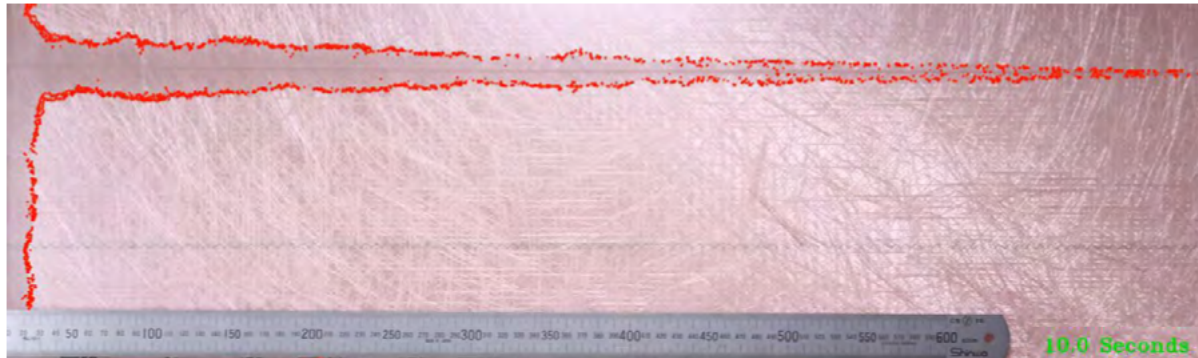


Figure 4. Example of the detected flow front at 10 seconds for a single channel flow experiment.

3.3. Experimental validation

To compare the experimental fluid transport performance of the different channel designs, the mean flow front position in the channel after 5, 7, and 9 seconds is shown relative to the equivalent permeability of each channel in Figure 5. Here the trend of increasing flow distance (or velocity) with increasing equivalent permeability is very clear for each time. Some scatter is present for each case, likely as a result of subtle differences in the bagging configuration at the inlet of each test. As expected, the reduction in width from reference Case A produces a significant reduction in equivalent permeability of Case B, as the cross-sectional area has not only reduced, but the aspect ratio has also increased. The increase in depth for Case C doubles the cross-sectional area at the cost of doubling the aspect ratio, which is seen experimentally to produce a modest increase in the equivalent permeability. These experimental observations agree well with the predictions from the parametric analysis in Section 2.2.

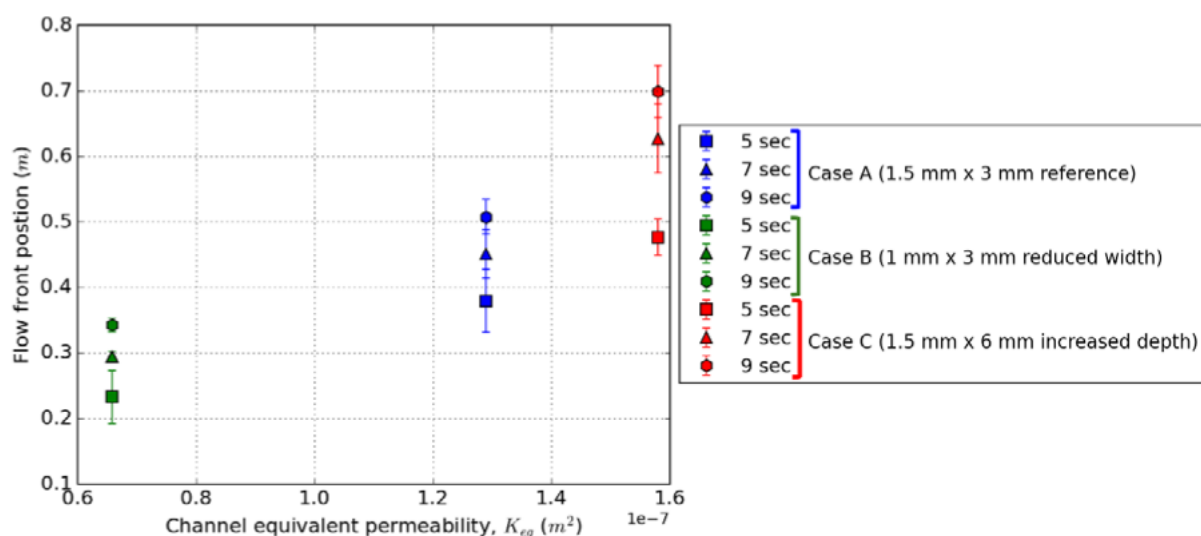


Figure 5. The flow front position relative to the effective permeability of the different channels, as measured at 5, 7 and 9 seconds.

4. Optimization of resin uptake

For comparing different vacuum infusion channel geometries in terms of resin uptake, a baseline 20 mm cut-to-cut spacing between channels has been chosen for a PET foam core material with 200 kg/m^3 density. The resin density is assumed to be 1150 kg/m^3 , representative of a typical two-part Epoxy system used for wind turbine blades. A resin uptake value of 0.415 kg/m^2 has been selected for this analysis based on publicly available data for a 200 kg/m^3 density PET foam [10]. Subsequently, for one surface of a core material with such a configuration of infusion channels, the total mass of resin (both filling the channels and uptake into the core surface pores) per square meter can be simply calculated. Based on the understanding that square channels (with an aspect ratio of 1) provide better flow performance relative to cross-sectional area, several square channel designs of similar width, a similar equivalent permeability, and a similar cross-sectional area to the reference Case A, $1.5 \text{ mm} \times 3 \text{ mm}$ channel, have been investigated to quantify any potential change in total resin uptake. The performance of these cases, in addition to rectangular cases of reduced width (Case B) or increased depth (Case C), are listed in Table 1. The same comparison of the different designs is also shown graphically in Figure 6.

Notably, due to the improved flow efficiency through a square channel, an 11.3% reduction (85 g/m^2) in the total resin mass per square meter can be achieved by reducing the wall dimensions and cross-sectional area without affecting the flow velocity. For the same cross-sectional area as the rectangular reference, a square design can still reduce the total resin mass by 4.8% while simultaneously increasing the flow velocity by 10.9%. Alternatively, at the cost of a 21.6% reduction in flow velocity, a square design of the same width as the reference can achieve a 22.5% reduction (169 g/m^2) in the total resin areal mass. The higher aspect ratio Cases (B and C) generally perform quite poorly relative to the reference Case A, and against the square channel cases in particular.

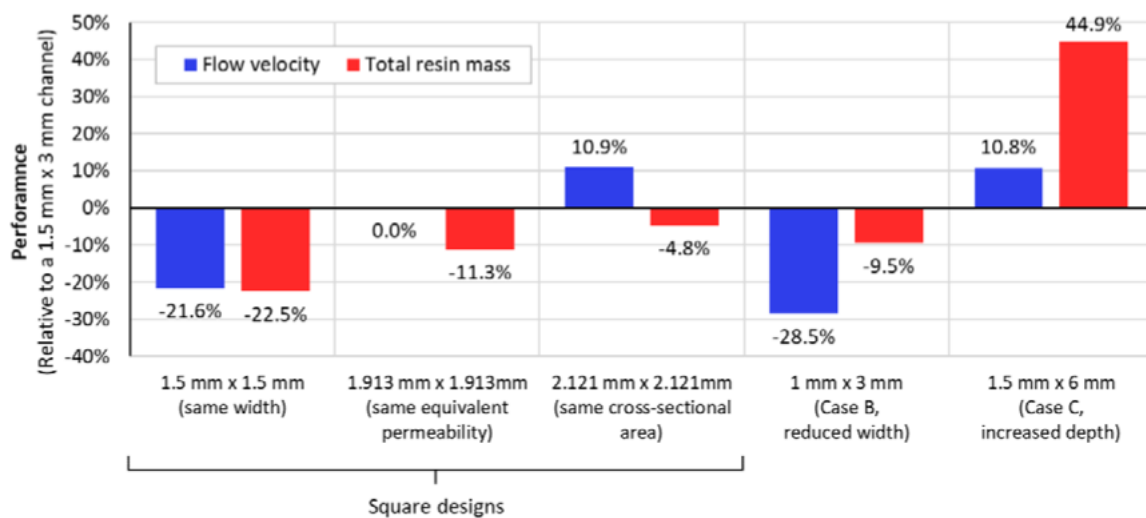


Figure 6. Comparison of alternative channel designs relative to a $1.5 \text{ mm} \times 3 \text{ mm}$ reference, in terms of resin flow velocity and total resin mass (combination of core material uptake and filled channels).

Table 1. Comparison of alternative channel designs relative to a 1.5 mm × 3mm reference.

Channel design	Aspect ratio	Equivalent permeability (m ²)	Relative flow velocity	Total resin areal mass (kg/m ²)	Relative total resin mass
1.5 mm × 3 mm Reference (Case A)	2	1.29×10 ⁻⁷	-	0.753	-
1.5 mm × 1.5 mm (same width)	1	7.91×10 ⁻⁸	-21.6%	0.584	-22.5%
1.913 mm × 1.913 mm (same equivalent permeability)	1	1.29×10 ⁻⁷	0.0%	0.668	-11.3%
2.121 mm × 2.121 mm (same cross-sectional area)	1	1.58×10 ⁻⁷	10.9%	0.717	-4.8%
1 mm × 3 mm (Case B, reduced width)	3	6.58×10 ⁻⁸	-28.5%	0.682	-9.5%
1.5 mm × 6 mm (Case C, increased depth)	4	1.58×10 ⁻⁷	10.8%	1.092	44.9%

5. Conclusions

This work demonstrated the importance of considering the vacuum infusion channel design in core materials for VARI processes. An analytical study of the relationship between the channel geometry and equivalent permeability revealed that rectangular channels have the most efficient performance in terms of equivalent permeability and fluid flow, when AR tends to 1 (a square cross-section). This dependence of equivalent permeability on channel geometry was validated with flow experiments through channels of different geometries on top of a stack of biaxial non-crimp glass-fibre fabric.

Additionally, by considering the effect of resin uptake with different channel designs, it was shown that the selection of a 1.913 mm × 1.913 mm square shape would lead to an 11.3% reduction in the total resin mass per square meter for the same equivalent permeability, compared against a common 1.5 mm × 3 mm channel design. Otherwise, for the same cross-sectional area, the selection of a square shape led to a 10.9% increase of the equivalent permeability and a 4.8% reduction in resin mass over the reference rectangular design.

Future work plans to extend the optimization of core channels to their cut-to-cut spacing. The effect of core groove spacing will be studied for different groove dimensions and fabric skins in order to achieve optimal resin utilization and minimize total resin uptake. The ultimate goal is to improve core material groove design and patterning guidelines.

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